WEEK 7 · OBJECT RECOGNITION

How many bikes are in the rack?

A photo in, a count out

Felix Reichle



What object recognition does

INPUT

One photo

OUTPUT

For every object it finds: a box, a label and a confidence between 0 and 1

USES

Counting traffic, checking shelf stock, spotting animals on wildlife cameras



From phone photo to model input

1. **Protect privacy:** blur people and a number plate, remove GPS data
2. **Open:** convert every photo to RGB
3. **Rotate:** apply the phone's stored rotation (EXIF)
4. **Resize:** longest side at most 1024 px
5. **Tensor:** pixels from 0–255 to 0–1, shape 3 × height × width
6. **Inside the model:** short side to 800 px, colours normalised with ImageNet values

Ground truth, made by hand

I counted the bikes on each photo before looking at any boxes: every foreground bike, also if only partly visible.

Then I marked the wrong boxes: no bike, a second box on one bike, or a background bike.

What I built

USER

A student who wants to know if there is a free spot before walking to the rack

MODEL

Faster R-CNN, pre-trained on COCO, which already includes “bicycle”. No training needed for an MVP.

PHOTOS

9 of my own, 83 bikes in total: nearly empty to packed, outdoors, indoors and at night

MVP

Count the “bicycle” boxes above a confidence threshold: a photo in, a number out







Evaluation: how far off is the number?

Main metric: count error (MAE). My user only sees the number, so what matters is how far it is off.

1.33
bikes off per photo, at threshold 0.85

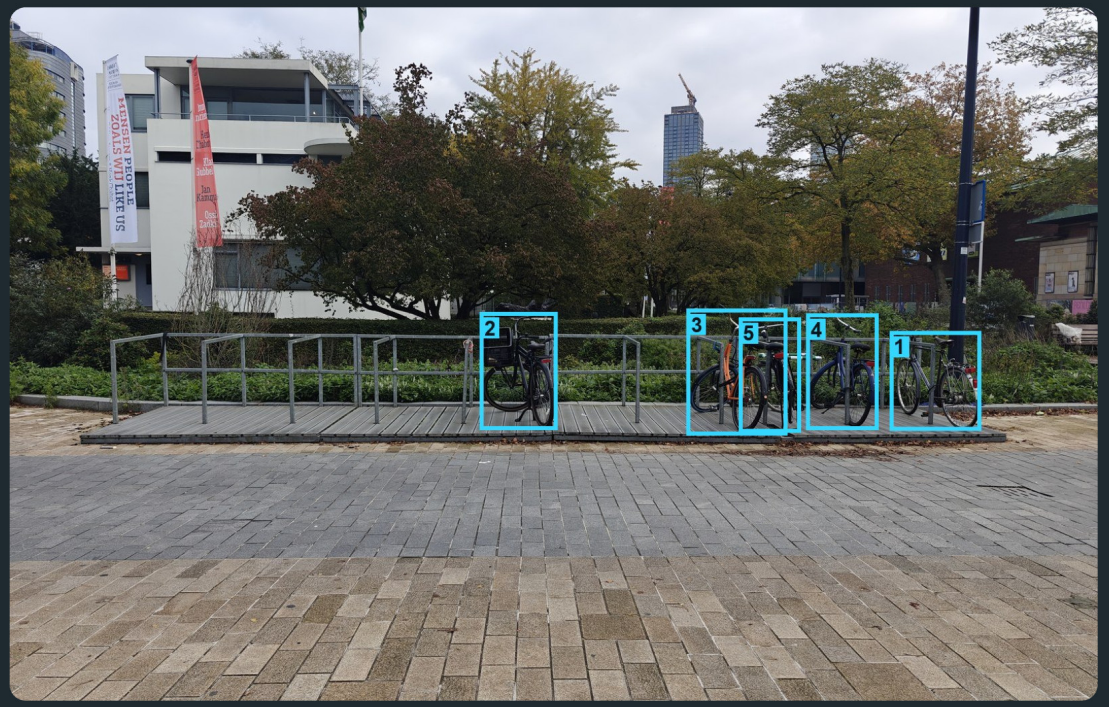
3 of 9
counts exactly right

0.83
precision: boxes that are right

0.83
recall: bikes that were found

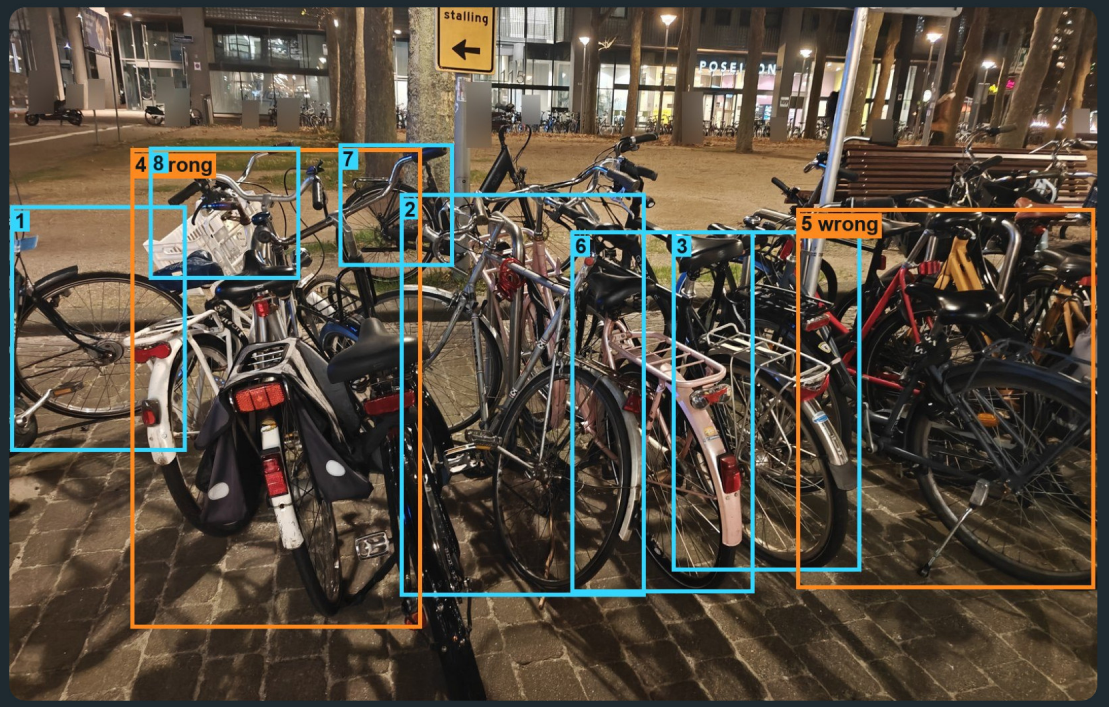


End result: where it works and where it fails



Works: 5 boxes, 5 bikes, all right

Few bikes, clearly apart, in daylight



Fails: 8 boxes, 13 bikes, 2 wrong

Overlapping bikes at night: only 6 of 13 found correctly

Who it could fail or harm

Students at busy racks or after dark	It is weakest exactly when they need it most. The harm is a wasted walk.
Owners of unusual bikes	I did not test cargo bikes or adapted bikes, so they may not be counted.
People walking past	A camera aimed at a rack also films them. I blurred people and a number plate in my photos.
Bike owners, if it is misused	With an error this size, the count should not decide things like removing bikes.

Next: add rack capacity to get real free spots, and test with more photos at night.